

Topic-Wise Question Bank

Organized from Easy to Difficult

Questions only - arranged topic-wise for revision and exam preparation.

TOTAL QUESTIONS	54
EASY	26
MEDIUM	18
DIFFICULT	10

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01. Algorithms, Search, Sorting & Graphs	10 questions
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TOPIC 01 - Algorithms, Search, Sorting & Graphs

10 questions - ordered from easier to harder within this topic.

EASY

Question 01. Compare BFS & DFS with examples.

Question 02. Describe step-by-step how Binary Search locates a target value in a sorted array. Why does it fail if the array is unsorted?

Question 03. Explain the logic of "Bubble Sort". Why is it considered inefficient for large datasets compared to Merge Sort?

Question 04. Explain the time complexity of merge sort. Best, Average, Worst.

MEDIUM

Question 05. Problem solved more efficiently in adjacency list representation than adjacency matrix representation and problem solved more effective in adjacency matrix adjacency list.

Question 06. Given an adjacency list representation for a complete binary tree on 7 vertices, give the equivalent adjacency matrix representation. Assume that vertices are numbered from 1 to 7 as in a binary heap.

Question 07. Solve Fibonacci series using dynamic programming.

Question 08. Construct a logical argument explaining why a heuristic (like A search) can be faster than a blind search (like BFS), even if it doesn't guarantee the absolute shortest path in all cases.

DIFFICULT

Question 09. How to Determine the weighted graph has negative cycle?

Question 10. X is an integer stream of N numbers. You have to select 2 data P and Q such that $A \leq (P+Q) \leq B$. Write an algorithm / pseudocode / C program to find how many ways you can select P & Q. The time complexity must be $O(N \log N)$.

TOPIC 02 - Data Structures

5 questions - ordered from easier to harder within this topic.

EASY

Question 01. Difference between Stack and Queue. Write about 2 problems solved by stack and queue.

Question 02. What is Hash function? Describe it.

Question 03. Linked list, doubly linked list and circular linked list explains with diagram.

MEDIUM

Question 04. You have two stacks. Explain the logic required to implement a Queue (FIFO) using only these two stacks.

DIFFICULT

Question 05. Consider a hash table of size 13 strong entities with integer keys. Suppose the hash function is $h(k) = k \bmod 13$. Insert in the given order entries with keys 10, 3, 6, 16, 17, 19, 10, 3, 6, 16, 17, 19 into the hash table using linear probing to resolve collisions. Show all the work.

TOPIC 03 - Programming, Flowcharts & Recursion

9 questions - ordered from easier to harder within this topic.

EASY

Question 01. Draw or describe a Flowchart to determine the largest of three distinct numbers (A, B, C).

Question 02. Write a function to find the smallest element from an array.

Question 03. Given two integers A and B as input, write a program to compute the least common multiple (LCM) of A and B.

Question 04. Write a Program Prime number print from 1 to n.

Question 05. Write a Program Floyd's triangle n=5:

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

Question 06. Write a C Program Find sum of the series: $1+2+4+7+11+\dots+N$

MEDIUM

Question 07. Write a code in C/C++ that will output the 2nd largest number. (If $N \geq 1$)

Question 08. What is Multithreading programming? Why is multithreading used in programming?

DIFFICULT

Question 09. Output find out from recursion:

```
#include <stdio.h>
void fun(int x) {
    if(x < 0) return;
    printf("%d\n", x--);
    fun(--x);
    printf("%d\n", x);
}
int main() {
    fun(5);
    return 0;
}
```

TOPIC 04 - Object-Oriented Programming & Java

3 questions - ordered from easier to harder within this topic.

EASY

Question 01. Explain the concept of Inheritance in Object-Oriented Programming. Name and briefly describe three common types of inheritance used in OOP.

Question 02. What is Polymorphism? Discuss about different types of Polymorphism with example?

DIFFICULT

Question 03. Consider the following Java code. Mention which of the methods overload, override, and hide superclass methods. What about the remaining method?

```
public class Class A {
    public void m1() {}
    public void m2(int i) {}
    public void m3(int i) {}
    public static void m4(int i) {}
}
public class Class B extends Class A {
    public static void m1(int i) {}
    public void m2(int i) {}
    public void m3(String s) {}
    public static void m4(int i) {}
}
```

TOPIC 05 - Database Management Systems & SQL

8 questions - ordered from easier to harder within this topic.

EASY

Question 01. What is the full meaning of SQL? List of the aggregate function. Write SQL Query of a table and its output.

Question 02. Discuss about different types of relations in DBMS.

Question 03. You are designing a database for a Library. Explain the relationship between a "Book" table and a "Borrower" table. Is it 1-to-1, 1-to-Many, or Many-to-Many? Justify your answer.

MEDIUM

Question 04. Assuming an Employees table with the structure: employee_id, employee_name, department_id, and salary. (a) To find the total number of employees in each department. (b) To calculate the average salary across all employees in the organization.

Question 05. Suppose that we have a relational database with the following tables. Underlined attributes represent primary keys: Movies (mid, title, year); People (pid, name); Genres (gid, genre); HasRole (pid, mid, role); HasGenre (gid, mid). Write an SQL query to return the number of movies that are romantic comedies.

Question 06. Draw the ER diagram of the Library management system.

Question 07. In a B+ tree, each node can have a maximum of 4 child pointers (order $m = 4$). From height 0 to height 2 (where height 0 is the root), what is the maximum number of keys that can be stored across all nodes in the tree?

DIFFICULT

Question 08. Consider a concurrent transaction schedule involving two data items, X and Y. T1: Read(X), Write(X), Read(Y), Write(Y). T2: Read(Y), Write(Y), Read(X), Write(X). Describe how the standard 2PL protocol treats these transactions. Contrast with Strict 2PL.

TOPIC 06 - Computer Networks & Data Communication

5 questions - ordered from easier to harder within this topic.

EASY

Question 01. Compare between TCP and UDP: their connection, reliability, speed. One real-life use case of UDP.

Question 02. Write down the functionality of OSI model.

Question 03. Explain the logic of a "Checksum". How is it used to verify data integrity during file transfer?

MEDIUM

Question 04. CRC is a redundancy error technique used to determine the error. Suppose the original data is 11100 and the divisor is 1001.

DIFFICULT

Question 05. Apply the concept of IP addressing and routing to explain how packets are delivered across networks using Open Shortest Path First (OSPF) at the network layer.

TOPIC 07 - Cybersecurity & Cryptography

6 questions - ordered from easier to harder within this topic.

EASY

Question 01. Difference between spoofing and sniffing.

MEDIUM

Question 02. Attacker steals private key of website that uses transport layer security and remains undetected. What can be done with a private key?

Question 03. Preserving confidentiality, integrity, and availability of data is a restatement of the concern over falsification, masquerade, and denial of service. Explain how the first three concepts relate to the last four.

Question 04. In order to prevent that, the company decided to add end-to-end encryption techniques. Which layer of the OSI model is suitable to work in, considering parameters like development time, software maintainability, and development cost? Give reasons for your concepts.

DIFFICULT

Question 05. Describe a man-in-the-middle attack on the Diffie-Hellman key exchange protocol in which the adversary generates two public key pairs for the attack.

Question 06. A & B two frames in a browser loaded from different origins. Why is it a reasonable security policy to allow A to navigate B to another origin based only on whether the display area of A contains dis-pare of B and A has the control over area.

TOPIC 08 - Software Engineering & Version Control

4 questions - ordered from easier to harder within this topic.

EASY

Question 01. Assume a banking application wants to add a fingerprint system for their App. Write one functional requirement and one security requirement.

Question 02. What is Version Control (e.g., Git)? Explain the specific difference between "Committing" code and "Pushing" code.

MEDIUM

Question 03. Which system did you build for a real-life software project? What problems you faced during that time and how to solve this?

DIFFICULT

Question 04. Given the following values, compute function point when all complexity adjustment factor (CAF) and weighting factors are average: User Input = 50; User Output = 40; User Inquiries = 35; User Files = 6; External Interface = 4.

TOPIC 09 - Machine Learning & Data Preprocessing

2 questions - ordered from easier to harder within this topic.

EASY

Question 01. Compare and contrast the three fundamental paradigms of Machine Learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

MEDIUM

Question 02. Suppose your dataset has missing values and noise. How would you preprocess it?

TOPIC 10 - Digital Logic

1 questions - ordered from easier to harder within this topic.

MEDIUM

Question 01. What is a Universal Gate? Prove that the NAND Gate is a Universal Gate.

TOPIC 11 - Mathematics

1 questions - ordered from easier to harder within this topic.

EASY

Question 01. A rectangle and a square have the same area. The square has a perimeter of 32 meters. The width of the rectangle is 4 meters. Find the length of the rectangle (in meters).